

Отчет по лабораторной работе №2 модуля 4

Тема: Настройка протокола STP (IEEE 802.1D)

Выполнила: Васильева Юлия Вячеславовна

1. Для заданной на схеме schema-lab2 сети, состоящей из управляемых коммутаторов и персональных компьютеров настроить протокол STP, назначив явно один из коммутаторов корневым настройкой приоритета.

Для Layer2Switch-1(корневой коммутатор):

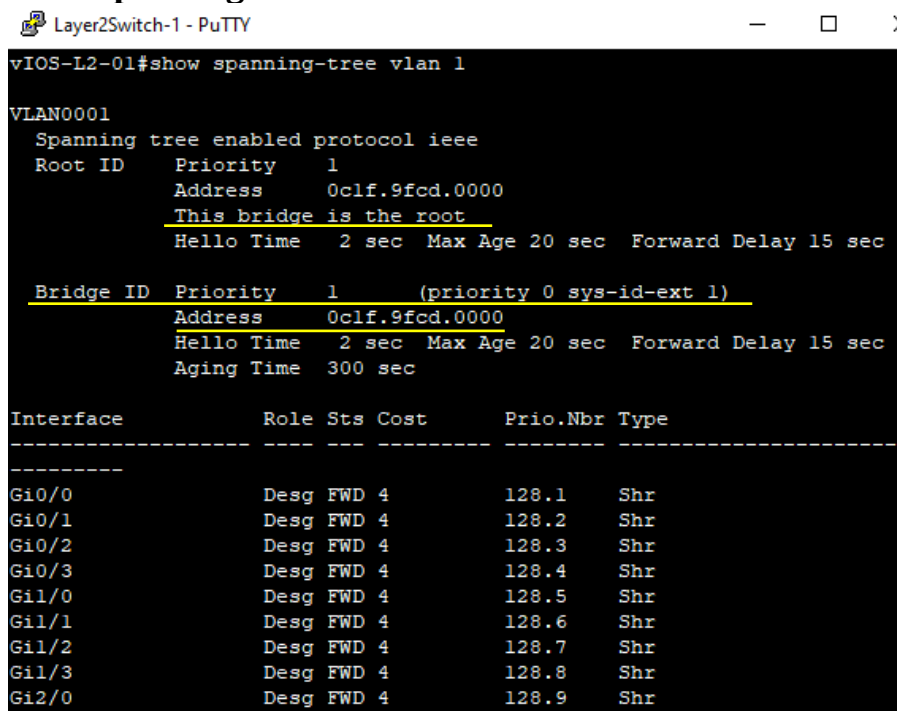
```
enable
configure terminal
spanning-tree mode pvst
spanning-tree vlan 1 priority 0
exit
write memory
```

Для коммутаторов с Layer2Switch-2 по Layer2Switch-5:

```
enable
configure terminal
spanning-tree mode pvst
spanning-tree vlan 1
exit
write memory
```

На коммутаторе1:

```
show spanning-tree vlan 1
```



```
Layer2Switch-1 - PuTTY
vIOS-L2-01#show spanning-tree vlan 1
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority      1
            Address      0c1f.9fcd.0000
            This bridge is the root
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority      1   (priority 0 sys-id-ext 1)
            Address      0c1f.9fcd.0000
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Desg FWD 4        128.1   Shr
Gi0/1                    Desg FWD 4        128.2   Shr
Gi0/2                    Desg FWD 4        128.3   Shr
Gi0/3                    Desg FWD 4        128.4   Shr
Gi1/0                    Desg FWD 4        128.5   Shr
Gi1/1                    Desg FWD 4        128.6   Shr
Gi1/2                    Desg FWD 4        128.7   Shr
Gi1/3                    Desg FWD 4        128.8   Shr
Gi2/0                    Desg FWD 4        128.9   Shr
```

На коммутаторе2:

show spanning-tree vlan 1

Layer2Switch-2 - PuTTY

```
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
            Address    0c1f.9fcd.0000
            Cost        4
            Port        1 (GigabitEthernet0/0)
            Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
            Address    0cd7.214e.0000
            Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time   300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Root FWD 4         128.1   Shr
Gi0/1                    Altn BLK 4         128.2   Shr
Gi0/2                    Altn BLK 4         128.3   Shr
Gi0/3                    Altn BLK 4         128.4   Shr
Gi1/0                    Altn BLK 4         128.5   Shr
Gi1/1                    Altn BLK 4         128.6   Shr
Gi1/2                    Altn BLK 4         128.7   Shr
Gi1/3                    Altn BLK 4         128.8   Shr
Gi2/0                    Desg FWD 4         128.9   Shr
```

На коммутаторе3:

show spanning-tree vlan 1

Layer2Switch-3 - PuTTY

```
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
            Address    0c1f.9fcd.0000
            Cost        4
            Port        1 (GigabitEthernet0/0)
            Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
            Address    0c86.931f.0000
            Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time   300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Root FWD 4         128.1   Shr
Gi0/1                    Altn BLK 4         128.2   Shr
Gi0/2                    Desg FWD 4         128.3   Shr
Gi0/3                    Desg FWD 4         128.4   Shr
Gi1/0                    Desg FWD 4         128.5   Shr
Gi1/1                    Desg FWD 4         128.6   Shr
```

На коммутаторе4:

show spanning-tree vlan 1

```
Layer2Switch-4 - PuTTY
disk successfully.
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0clf.9fcd.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32769 (priority 32768 sys-id-ext 1)
             Address     0c4b.99b4.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time   300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Root FWD 4         128.1   Shr
Gi0/1                    Altn BLK 4         128.2   Shr
Gi0/2                    Desg FWD 4         128.3   Shr
Gi0/3                    Desg FWD 4         128.4   Shr
Gil/0                    Desg FWD 4         128.5   Shr
Gil/1                    Desg FWD 4         128.6   Shr
```

На коммутаторе5:

show spanning-tree vlan 1

```
Layer2Switch-5 - PuTTY
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0clf.9fcd.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32769 (priority 32768 sys-id-ext 1)
             Address     0cbf.cbfe.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time   300 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Root FWD 4         128.1   Shr
Gi0/1                    Altn BLK 4         128.2   Shr
Gi0/2                    Desg FWD 4         128.3   Shr
Gi0/3                    Desg FWD 4         128.4   Shr
Gil/0                    Desg FWD 4         128.5   Shr
Gil/1                    Desg FWD 4         128.6   Shr
```

2. Проверить доступность каждого с каждым всех персональных компьютеров (VPCS), результаты запротоколировать.

С ПК1:

ip 200.100.2.1/24

save

ping 200.100.2.2 -c 1

ping 200.100.2.3 -c 1

ping 200.100.2.4 -c 1

ping 200.100.2.5 -c 1

ping 200.100.2.6 -c 1

```
PC1> ping 200.100.2.2 -c 1
84 bytes from 200.100.2.2 icmp_seq=1 ttl=64 time=11.235 ms
PC1> ping 200.100.2.3 -c 1
84 bytes from 200.100.2.3 icmp_seq=1 ttl=64 time=27.284 ms
PC1> ping 200.100.2.4 -c 1
84 bytes from 200.100.2.4 icmp_seq=1 ttl=64 time=15.809 ms
PC1> ping 200.100.2.5 -c 1
84 bytes from 200.100.2.5 icmp_seq=1 ttl=64 time=13.412 ms
PC1> ping 200.100.2.6 -c 1
84 bytes from 200.100.2.6 icmp_seq=1 ttl=64 time=12.893 ms
```

С ПК2:

ip 200.100.2.2/24

save

ping 200.100.2.1 -c 1

ping 200.100.2.3 -c 1

ping 200.100.2.4 -c 1

ping 200.100.2.5 -c 1

ping 200.100.2.6 -c 1

```
PC2> ping 200.100.2.1 -c 1

84 bytes from 200.100.2.1 icmp_seq=1 ttl=64 time=11.266 ms

PC2> ping 200.100.2.3 -c 1

84 bytes from 200.100.2.3 icmp_seq=1 ttl=64 time=12.567 ms

PC2> ping 200.100.2.4 -c 1

84 bytes from 200.100.2.4 icmp_seq=1 ttl=64 time=17.234 ms

PC2> ping 200.100.2.5 -c 1

84 bytes from 200.100.2.5 icmp_seq=1 ttl=64 time=12.184 ms

PC2> ping 200.100.2.6 -c 1

84 bytes from 200.100.2.6 icmp_seq=1 ttl=64 time=34.783 ms
```

С ПК3:

ip 200.100.2.3/24

save

ping 200.100.2.1 -c 1

ping 200.100.2.2 -c 1

ping 200.100.2.4 -c 1

ping 200.100.2.5 -c 1

ping 200.100.2.6 -c 1

```
PC3> ping 200.100.2.1 -c 1

84 bytes from 200.100.2.1 icmp_seq=1 ttl=64 time=11.526 ms

PC3> ping 200.100.2.2 -c 1

84 bytes from 200.100.2.2 icmp_seq=1 ttl=64 time=28.891 ms

PC3> ping 200.100.2.4 -c 1

84 bytes from 200.100.2.4 icmp_seq=1 ttl=64 time=3.695 ms

PC3> ping 200.100.2.5 -c 1

84 bytes from 200.100.2.5 icmp_seq=1 ttl=64 time=29.238 ms

PC3> ping 200.100.2.6 -c 1

84 bytes from 200.100.2.6 icmp_seq=1 ttl=64 time=14.046 ms
```

C ПК4:

ip 200.100.2.4/24

save

ping 200.100.2.1 -c 1

ping 200.100.2.2 -c 1

ping 200.100.2.3 -c 1

ping 200.100.2.5 -c 1

ping 200.100.2.6 -c 1

```
PC4> ping 200.100.2.1 -c 1
84 bytes from 200.100.2.1 icmp_seq=1 ttl=64 time=20.111 ms
PC4> ping 200.100.2.2 -c 1
84 bytes from 200.100.2.2 icmp_seq=1 ttl=64 time=17.514 ms
PC4> ping 200.100.2.3 -c 1
84 bytes from 200.100.2.3 icmp_seq=1 ttl=64 time=4.404 ms
PC4> ping 200.100.2.5 -c 1
84 bytes from 200.100.2.5 icmp_seq=1 ttl=64 time=20.932 ms
PC4> ping 200.100.2.6 -c 1
84 bytes from 200.100.2.6 icmp_seq=1 ttl=64 time=24.751 ms
```

C ПК5:

ip 200.100.2.5/24

save

ping 200.100.2.1 -c 1

ping 200.100.2.2 -c 1

ping 200.100.2.3 -c 1

ping 200.100.2.4 -c 1

ping 200.100.2.6 -c 1

```
PC5> ping 200.100.2.1 -c 1

84 bytes from 200.100.2.1 icmp_seq=1 ttl=64 time=19.041 ms

PC5> ping 200.100.2.2 -c 1

84 bytes from 200.100.2.2 icmp_seq=1 ttl=64 time=17.458 ms

PC5> ping 200.100.2.3 -c 1

84 bytes from 200.100.2.3 icmp_seq=1 ttl=64 time=19.157 ms

PC5> ping 200.100.2.4 -c 1

84 bytes from 200.100.2.4 icmp_seq=1 ttl=64 time=14.635 ms

PC5> ping 200.100.2.6 -c 1

84 bytes from 200.100.2.6 icmp_seq=1 ttl=64 time=3.265 ms
```

C ПК6:

ip 200.100.2.6/24

save

ping 200.100.2.1 -c 1

ping 200.100.2.2 -c 1

ping 200.100.2.3 -c 1

ping 200.100.2.4 -c 1

ping 200.100.2.5 -c 1

```
PC6> ping 200.100.2.1 -c 1

84 bytes from 200.100.2.1 icmp_seq=1 ttl=64 time=23.282 ms

PC6> ping 200.100.2.2 -c 1

84 bytes from 200.100.2.2 icmp_seq=1 ttl=64 time=21.735 ms

PC6> ping 200.100.2.3 -c 1

84 bytes from 200.100.2.3 icmp_seq=1 ttl=64 time=12.270 ms

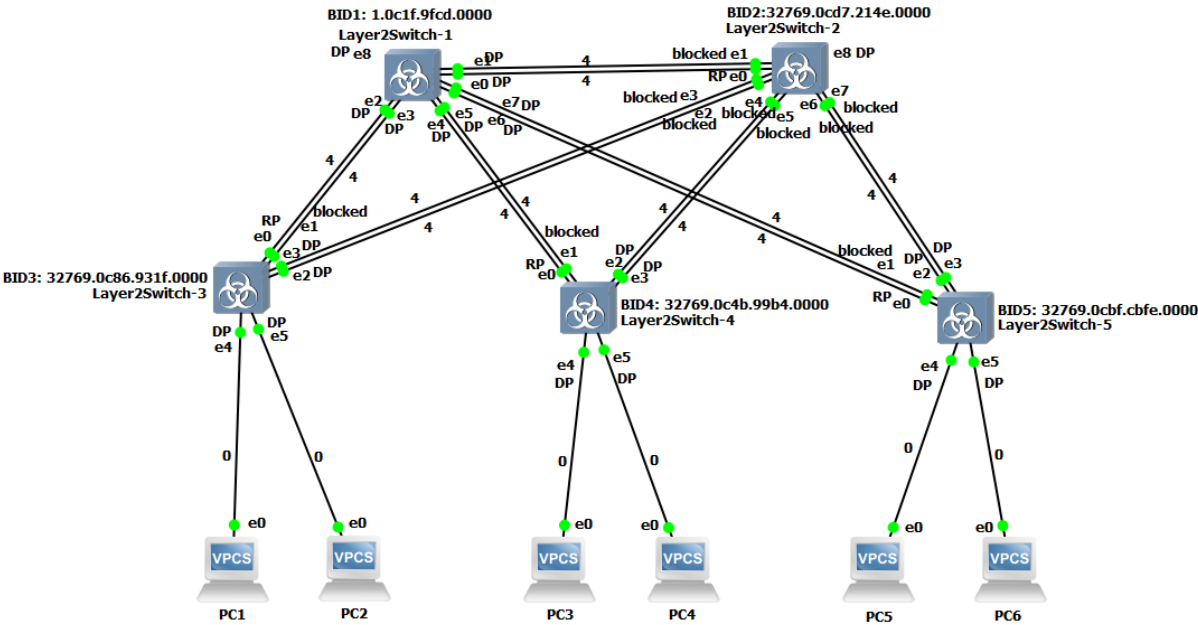
PC6> ping 200.100.2.4 -c 1

84 bytes from 200.100.2.4 icmp_seq=1 ttl=64 time=13.639 ms

PC6> ping 200.100.2.5 -c 1

84 bytes from 200.100.2.5 icmp_seq=1 ttl=64 time=5.356 ms
```

3. На изображении схемы отметить BID каждого коммутатора и режимы работы портов (RP/DP/blocked) и стоимости маршрутов, результат сохранить в файл.



4. При помощи wireshark отследить передачу пакетов hello от корневого коммутатора на всех линках (nb!), результаты включить в отчет.

0c1f.9fcd.0000

Коммутатор1:

Линк 1(к коммутатору2):

*Standard input [Layer2Switch-1 Ethernet0 to Layer2Switch-2 Ethernet0]

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:d7:21:4e:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:d7:21:4e:00:00 Cost = 0 Port = 0x8001
2	1.268096	0c:1f:9f:cd:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8001
3	1.491336	0c:1f:9f:cd:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8001
4	2.269139	0c:1f:9f:cd:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8001
5	2.485244	0c:d7:21:4e:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:d7:21:4e:00:00 Cost = 0 Port = 0x8001
6	2.495674	0c:d7:21:4e:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:d7:21:4e:00:00 Cost = 0 Port = 0x8001

Линк 2(к коммутатору2):

*Standard input [Layer2Switch-1 Ethernet1 to Layer2Switch-2 Ethernet1]

No.	Time	Source	Destination	Protocol	Length	Info
5	0.246439	0c:d7:21:4e:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:d7:21:4e:00:00 Cost = 0 Port = 0x8002
7	0.390602	0c:1f:9f:cd:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8002
8	0.397866	0c:1f:9f:cd:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8002
9	1.926747	0c:1f:9f:cd:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8002
10	2.639000	0c:1f:9f:cd:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8002
11	2.892620	0c:d7:21:4e:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:d7:21:4e:00:00 Cost = 0 Port = 0x8002

Линк 3(к коммутатору3):

*Standard input [Layer2Switch-1 Ethernet2 to Layer2Switch-3 Ethernet0]

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:1f:9f:cd:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8003
2	0.415841	0c:1f:9f:cd:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8003
3	0.416892	0c:1f:9f:cd:00:02	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8003
4	0.427855	0c:86:93:1f:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:86:93:1f:00:00 Cost = 0 Port = 0x8001
5	0.946587	0c:86:93:1f:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:86:93:1f:00:00 Cost = 0 Port = 0x8001
6	1.067494	0c:86:93:1f:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:86:93:1f:00:00 Cost = 0 Port = 0x8001

Линк 4(к коммутатору3):

*Standard input [Layer2Switch-1 Ethernet3 to Layer2Switch-3 Ethernet1]

No.	Time	Source	Destination	Protocol	Length	Info
4	0.796258	0c:1f:9f:cd:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8004
5	1.060109	0c:86:93:1f:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:86:93:1f:00:00 Cost = 0 Port = 0x8002
6	1.138759	0c:1f:9f:cd:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8004
7	1.177584	0c:86:93:1f:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:86:93:1f:00:00 Cost = 0 Port = 0x8002
9	2.162811	0c:1f:9f:cd:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8004
10	2.734548	0c:1f:9f:cd:00:03	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8004

Линк 5(к коммутатору4):

*Standard input [Layer2Switch-1 Ethernet4 to Layer2Switch-4 Ethernet0]

No.	Time	Source	Destination	Protocol	Length	Info
4	0.815767	0c:1f:9f:cd:00:04	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8005
5	1.204770	0c:1f:9f:cd:00:04	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8005
6	1.220898	0c:4b:99:b4:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:4b:99:b4:00:00 Cost = 0 Port = 0x8001
7	1.565454	0c:4b:99:b4:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:4b:99:b4:00:00 Cost = 0 Port = 0x8001
8	2.328406	0c:1f:9f:cd:00:04	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8005
9	2.955520	0c:1f:9f:cd:00:04	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8005

Линк 6(к коммутатору4):

*Standard input [Layer2Switch-1 Ethernet5 to Layer2Switch-4 Ethernet1]

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:1f:9f:cd:00:05	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8006
2	0.020494	0c:1f:9f:cd:00:05	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8006
3	0.072485	0c:4b:99:b4:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:4b:99:b4:00:00 Cost = 0 Port = 0x8002
4	0.356053	0c:1f:9f:cd:00:05	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8006
5	1.264817	0c:1f:9f:cd:00:05	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8006
6	1.708835	0c:4b:99:b4:00:01	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:4b:99:b4:00:00 Cost = 0 Port = 0x8002

Линк 7(к коммутатору5):

*Standard input [Layer2Switch-1 Ethernet6 to Layer2Switch-5 Ethernet0]

No.	Time	Source	Destination	Protocol	Length	Info
5	1.221880	0c:1f:9f:cd:00:06	Nearest-Customer-Br...	STP	60 Conf.	Root = 0/1/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8007
6	1.721271	0c:1f:9f:cd:00:06	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8007
7	2.282165	0c:bf:cb:fe:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/100/0c:bf:cb:fe:00:00 Cost = 0 Port = 0x8001
8	2.476590	0c:1f:9f:cd:00:06	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8007
9	2.650943	0c:bf:cb:fe:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/200/0c:bf:cb:fe:00:00 Cost = 0 Port = 0x8001
10	2.659516	0c:bf:cb:fe:00:00	Nearest-Customer-Br...	STP	60 Conf.	Root = 32768/300/0c:bf:cb:fe:00:00 Cost = 0 Port = 0x8001

Линк 8(к коммутатору5):

*Standard input [Layer2Switch-1 Ethernet7 to Layer2Switch-5 Ethernet1]

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:1f:9f:cd:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8008
2	0.013563	0c:bf:cb:fe:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:bf:cb:fe:00:00 Cost = 0 Port = 0x8002
3	0.015003	0c:1f:9f:cd:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8008
4	0.408403	0c:bf:cb:fe:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:bf:cb:fe:00:00 Cost = 0 Port = 0x8002
5	0.413745	0c:bf:cb:fe:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:bf:cb:fe:00:00 Cost = 0 Port = 0x8002
6	0.681589	0c:1f:9f:cd:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:1f:9f:cd:00:00 Cost = 0 Port = 0x8008

Коммутатор2: Линк 9(к коммутатору3):

*Standard input [Layer2Switch-3 Ethernet2 to Layer2Switch-2 Ethernet2]

No.	Time	Source	Destination	Protocol	Length	Info
4	0.122786	0c:d7:21:4e:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:d7:21:4e:00:00 Cost = 0 Port = 0x8003
6	0.713954	0c:86:93:1f:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:1f:9f:cd:00:00 Cost = 4 Port = 0x8003
7	1.767558	0c:86:93:1f:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:86:93:1f:00:00 Cost = 0 Port = 0x8003
8	1.980652	0c:86:93:1f:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:1f:9f:cd:00:00 Cost = 4 Port = 0x8003
9	2.325641	0c:86:93:1f:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:86:93:1f:00:00 Cost = 0 Port = 0x8003
10	2.429781	0c:86:93:1f:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:86:93:1f:00:00 Cost = 0 Port = 0x8003

Линк 10(к коммутатору3):

*Standard input [Layer2Switch-3 Ethernet3 to Layer2Switch-2 Ethernet3]

No.	Time	Source	Destination	Protocol	Length	Info
5	0.705841	0c:d7:21:4e:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:d7:21:4e:00:00 Cost = 0 Port = 0x8004
6	0.820218	0c:86:93:1f:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:1f:9f:cd:00:00 Cost = 4 Port = 0x8004
7	1.783260	0c:86:93:1f:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:86:93:1f:00:00 Cost = 0 Port = 0x8004
8	1.984494	0c:86:93:1f:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:1f:9f:cd:00:00 Cost = 4 Port = 0x8004
9	2.336288	0c:86:93:1f:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:86:93:1f:00:00 Cost = 0 Port = 0x8004
10	2.457653	0c:86:93:1f:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:86:93:1f:00:00 Cost = 0 Port = 0x8004

Линк 11(к коммутатору4):

*Standard input [Layer2Switch-2 Ethernet4 to Layer2Switch-4 Ethernet2]

No.	Time	Source	Destination	Protocol	Length	Info
4	0.716149	0c:4b:99:b4:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:1f:9f:cd:00:00 Cost = 4 Port = 0x8003
5	0.995023	0c:4b:99:b4:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:4b:99:b4:00:00 Cost = 0 Port = 0x8003
6	1.372451	0c:4b:99:b4:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:4b:99:b4:00:00 Cost = 0 Port = 0x8003
7	1.682707	0c:4b:99:b4:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:4b:99:b4:00:00 Cost = 0 Port = 0x8003
8	1.887979	0c:4b:99:b4:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:1f:9f:cd:00:00 Cost = 4 Port = 0x8003
9	2.703923	0c:d7:21:4e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:d7:21:4e:00:00 Cost = 0 Port = 0x8005

Линк 12(к коммутатору4):

*Standard input [Layer2Switch-2 Ethernet5 to Layer2Switch-4 Ethernet3]

No.	Time	Source	Destination	Protocol	Length	Info
4	0.168059	0c:4b:99:b4:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:1f:9f:cd:00:00 Cost = 4 Port = 0x8004
5	0.393188	0c:4b:99:b4:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:4b:99:b4:00:00 Cost = 0 Port = 0x8004
6	0.840552	0c:4b:99:b4:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:4b:99:b4:00:00 Cost = 0 Port = 0x8004
7	1.204793	0c:4b:99:b4:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:4b:99:b4:00:00 Cost = 0 Port = 0x8004
8	1.430130	0c:4b:99:b4:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:1f:9f:cd:00:00 Cost = 4 Port = 0x8004
11	2.545086	0c:d7:21:4e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:d7:21:4e:00:00 Cost = 0 Port = 0x8006

Линк 13(к коммутатору5):

*Standard input [Layer2Switch-2 Ethernet6 to Layer2Switch-5 Ethernet2]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:bf:cb:fe:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:1f:9f:cd:00:00 Cost = 4 Port = 0x8003
2	0.975050	0c:bf:cb:fe:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:bf:cb:fe:00:00 Cost = 0 Port = 0x8003
4	1.295957	0c:bf:cb:fe:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:1f:9f:cd:00:00 Cost = 4 Port = 0x8003
5	1.366139	0c:bf:cb:fe:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:bf:cb:fe:00:00 Cost = 0 Port = 0x8003
6	1.371292	0c:bf:cb:fe:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:bf:cb:fe:00:00 Cost = 0 Port = 0x8003
7	1.753259	0c:d7:21:4e:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:d7:21:4e:00:00 Cost = 0 Port = 0x8007

Линк 14(к коммутатору5):

*Standard input [Layer2Switch-2 Ethernet7 to Layer2Switch-5 Ethernet3]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:bf:cb:fe:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:bf:cb:fe:00:00 Cost = 0 Port = 0x8004
2	0.310661	0c:bf:cb:fe:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 0/1/0c:1f:9f:cd:00:00 Cost = 4 Port = 0x8004
3	0.385889	0c:bf:cb:fe:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:bf:cb:fe:00:00 Cost = 0 Port = 0x8004
4	0.393022	0c:bf:cb:fe:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:bf:cb:fe:00:00 Cost = 0 Port = 0x8004
5	0.966730	0c:d7:21:4e:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:d7:21:4e:00:00 Cost = 0 Port = 0x8008
6	0.970824	0c:d7:21:4e:00:07	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:d7:21:4e:00:00 Cost = 0 Port = 0x8008

5. Изменить стоимость маршрута для порта RP произвольного назначенного (designated) коммутатора, повторить действия из п.3, результат сохранить в отдельный файл.

На коммутаторе 2:

enable

configure terminal

interface gigabitethernet 0/0

spanning-tree cost 100

exit

exit

write memory

show spanning-tree vlan 1

```

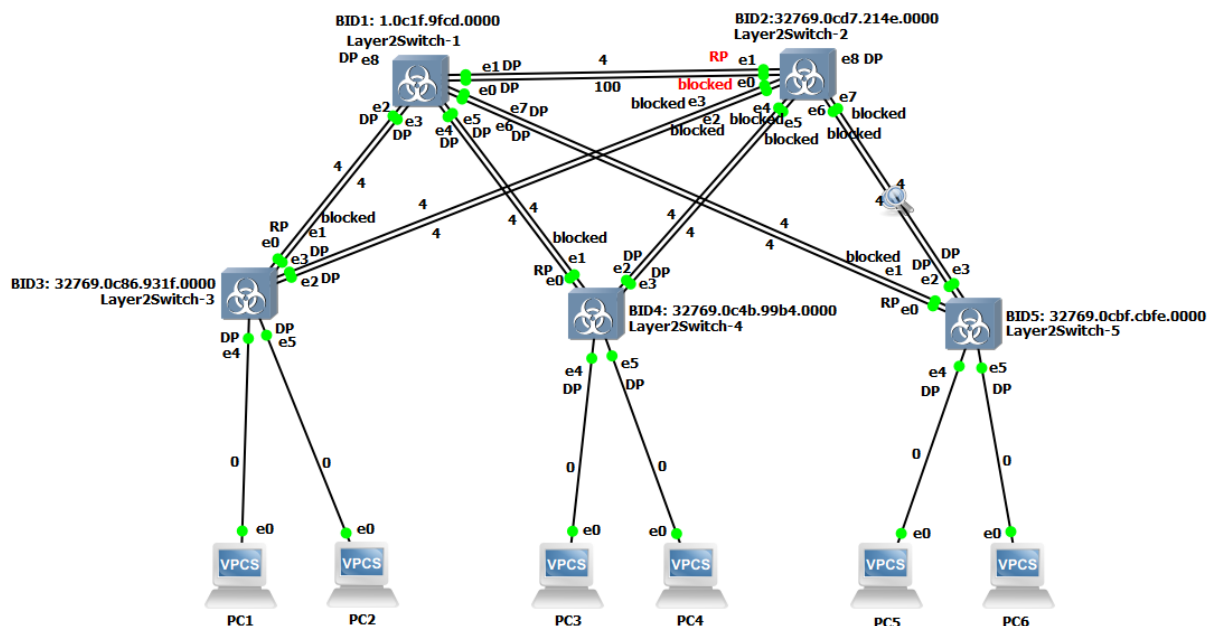
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0c1f.9fcd.0000
             Cost        4
             Port        2 (GigabitEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32769 (priority 32768 sys-id-ext 1)
             Address     0cd7.214e.0000
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  15 sec

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi0/0          Altn BLK 100      128.1   Shr
Gi0/1          Root FWD 4         128.2   Shr
Gi0/2          Altn BLK 4         128.3   Shr
Gi0/3          Altn BLK 4         128.4   Shr
Gi1/0          Altn BLK 4         128.5   Shr
Gi1/1          Altn BLK 4         128.6   Shr
Gi1/2          Altn BLK 4         128.7   Shr
Gi1/3          Altn BLK 4         128.8   Shr
Gi2/0          Desg FWD 4         128.9   Shr

```



6. Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств.

На каждом коммутаторе были проделанный действия, показанные в файле save_l2switch_config.pdf:

enable

term len 0

sh run

term len 25

disable